Reinforcement Learning for Adaptive Humanoid Stair Climbing

ME5418 Machine Learning in Robotics – Group 27

1. Motivation and Background

Many humanoid service robots operate in multi-level workspaces such as large shopping malls or households. When robots need to perform tasks across different floors, stair- climbing scenarios are frequently encountered. However, staircase geometries can vary considerably across different environments, including variations in step height, tread depth, and climbing direction, as in straight or turning staircases. Therefore, a practical humanoid robot should not only be capable of climbing a specific staircase, but should also be able to adapt its locomotion behavior to different staircase geometries.

Conventional humanoid stair-climbing methods typically rely on predefined motion trajectories or model-based planning and control. Pre-set joint trajectories are simple and efficient, but are difficult to adapt to variations in stair geometry. Foot-trajectory planning combined with inverse kinematics (IK) can generate joint configurations for a desired foot motion; however, the target foot placements and trajectories must first be specified according to the stair geometry, and kinematic feasibility alone does not guarantee dynamic stability. ZMP-based methods can further incorporate balance requirements but still rely on planned foot placements and reference trajectories [1]. Model predictive control (MPC) can explicitly consider robot dynamics and contact constraints, but requires an accurate model and may be computationally expensive for contact-rich motions. Consequently, changes in stair height, tread depth, or climbing direction generally require replanning or adjustment of the reference motion.

In contrast, reinforcement learning (RL) can learn a feedback policy that directly maps the observed robot state and stair geometry to joint commands. By providing geometric information such as stair height, tread depth, relative distance, and orientation as policy observations, the robot can adjust its foot clearance, placement, and body motion according to the encountered stair, without prescribing a complete reference trajectory for each staircase. This project therefore investigates whether a single RL policy can reduce the dependence on manually designed trajectories and achieve stable stair climbing across different staircase geometries.

2. Problem Statement

This project first considers a single-step stair-climbing task as the basic unit of staircase locomotion. At the beginning of each episode, the height and tread depth of the target step are randomly generated within predefined feasible ranges. The relative position and orientation between the robot and the target step can also be varied such that the robot must adjust its motion according to the current staircase geometry rather than execute a fixed joint trajectory.

The desired climbing behavior follows a *step-to* pattern. Starting from a stable standing configuration in front of the step, the robot first lifts one leg as the leading leg. The swing foot must obtain sufficient vertical clearance to avoid collision with the front face or edge of the step and then land within a valid support region on the upper surface. After the leading foot establishes stable contact, the trailing leg is lifted and brought onto the same step. The robot must subsequently transfer its body onto the step and recover a stable two-foot standing configuration.

The task is considered successfully completed when the robot's center-of-mass projection lies within the valid support region of the step, the robot remains upright on the step for a predefined stabilization period T_{stable}, and its heading is aligned with the desired direction toward the subsequent step. Throughout the climbing process, the robot should avoid falling, excessive body rotation, and undesired collisions between the legs and the vertical face or edge of the step. This single-step task provides the fundamental climbing behavior required for subsequent extension to multi-step staircases with varying heights, tread depths, and directions.

3. Reinforcement Learning Formulation

The stair-climbing task is formulated as a continuous-control RL problem. At each control step, the policy observes the target-step geometry and the robot state, and outputs commands for the lower-body joints.

- **Observation space:** The observation vector contains:
  - *Stair geometry:* target-step height h_s, tread depth w_s, distance to the next stair edge d_s, and heading error e_ψ relative to the desired climbing direction.
  - *Robot state:* joint positions $\mathbf{q}$, body angular velocity $\boldsymbol{\omega}$, projected gravity $\mathbf{g}_b \in \mathbb{R}^3$, left–right foot height difference $\Delta h_{LR} = z_L - z_R$ (positive when the left foot is higher), relative step length l_{step}, and the previous action $\mathbf{a}_{t-1}$.
- **Action space:** Continuous position commands for the actuated lower-body joints of the Unitree G1.
- **Reward structure:** The reward encourages the complete climbing sequence rather than only upward motion. Positive rewards are given for: (1) lifting the swing foot to approximately the target-step height plus a small

clearance margin; (2) placing the foot within a safe region on the upper surface of the step; (3) increasing torso height as the body moves onto the step; (4) aligning the robot with the desired direction; and (5) remaining upright and stable. Penalties are applied for collisions with the stair edge or vertical face and excessive body angular velocity.

- **Success and termination:** A trial is successful when both legs have completed the step-to climbing motion, the robot's center-of-mass projection lies within the valid support region of the target step, and the robot remains upright for a predefined stabilization period T_{stable}. The episode terminates if the robot falls, experiences severe undesired collision, or exceeds the time limit.

- **Neural network structure:** A two-branch MLP architecture is adopted for feature extraction. Robot joint states, including joint positions and angular velocities, are encoded by one MLP, while stair geometry and other scalar observations are encoded by a second MLP. The extracted features are concatenated and passed through fully connected layers for feature fusion. The fused representation is then used by the Actor and Critic to generate the continuous joint-action distribution and estimate the state value, respectively.

4. Learning Algorithm and Expected Results

Proximal Policy Optimization (PPO) will be adopted as the primary reinforcement learning algorithm. PPO is suitable for this task because humanoid stair climbing involves continuous multi-joint control and nonlinear foot–environment contact dynamics. Its clipped surrogate objective constrains excessively large policy updates and provides a relatively stable optimization procedure.

Training will be performed using multiple parallel simulation environments. During training, the target-step height, tread depth, initial distance, and relative orientation will be randomized within predefined feasible ranges. Consequently, the same policy is required to generate different lower-body motions according to the observed stair geometry rather than memorize a single fixed climbing trajectory.

The expected result is a single policy capable of completing the entire single-step climbing sequence:

$$\text{leading-leg lift} \rightarrow \text{safe foot placement} \rightarrow \text{trailing-leg following} \rightarrow \text{body elevation} \rightarrow \text{stable standing.} \tag{1}$$

More importantly, the learned policy is expected to adapt its foot clearance, placement, and body motion according to variations in step height, tread depth, and relative orientation.

The primary evaluation metric will be the **task success rate**, defined as the percentage of trials in which the robot reaches the target step. Additional evaluation metrics will include:

- **Collision rate**: frequency of undesired collisions with the stair edge or vertical face;
- **Fall rate**: percentage of episodes terminated because of loss of balance;
- **Completion time**: time required to complete one stair transition.

To evaluate the adaptability of the learned policy, a primary comparative experiment will be conducted between a policy trained using a **fixed staircase geometry** and a policy trained using **randomized staircase geometries**. Both policies will subsequently be evaluated over multiple staircase configurations, including geometries not encountered during training. This comparison will determine whether geometric randomization improves the ability of the policy to generalize beyond a specific staircase.

Overall, the project aims to demonstrate that PPO can learn a geometry-conditioned humanoid stair-climbing policy that maps the observed stair geometry and robot state directly to lower-body joint actions. The expected advantage is a reduced dependence on manually designed reference trajectories and improved adaptability to different staircase configurations.

References

[1] S. Kajita, F. Kanehiro, K. Kaneko, K. Fujiwara, K. Harada, K. Yokoi, and H. Hirukawa, "Biped Walking Pattern Generation by Using Preview Control of Zero-Moment Point," *Proceedings of the 2003 IEEE International Conference on Robotics and Automation (ICRA)*, vol. 2, pp. 1620–1626, 2003. doi: 10.1109/ROBOT.2003.1241826.